

## HAAST, New Zealand

WMO: 937090

Lat: 43.8604S

Lon: 169.0070E

Elev: 18

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>7      | (b)<br>34.4 | (c)<br>36.2 | (d)<br>18.3                   | (e)<br>13.8 | (f)<br>46.3 | (g)<br>22.8 | (h)<br>17.2 | (i)<br>45.9 | (j)<br>26.8           | (k)<br>49.9 | (l)<br>23.7 | (m)<br>52.2 | (n)<br>4.3            | (o)<br>150 | (p)<br>0.547 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>2      | (b)<br>12.4            | (c)<br>69.1     | (d)<br>63.0 | (e)<br>67.5 | (f)<br>62.1 | (g)<br>66.2 | (h)<br>61.4 | (i)<br>65.1         | (j)<br>67.2 | (k)<br>63.9 | (l)<br>66.1 | (m)<br>62.7 | (n)<br>65.1 | (o)<br>8.2           | (p)<br>280 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.3                            | 90.2 | 66.1 | 62.9 | 85.9 | 65.2 | 61.7 | 82.1 | 64.2 | 30.1          | 67.5 | 29.2 | 66.2 | 28.3 | 65.2 | 70.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 25.0          | (b) 21.4 | (c) 18.4 | (e) 30.1                   | (f) 73.2 | (g) 1.9            | (h) 2.6 | (i) 28.7                                           | (j) 75.0 | (k) 27.6   | (l) 76.5 | (m) 26.6   | (n) 78.0 | (o) 25.2   | (p) 79.8 |    |    |
| (a) 25.0          | (b) 21.4 | (c) 18.4 | (e) 29.0                   | (f) 66.5 | (g) 2.0            | (h) 1.6 | (i) 27.5                                           | (j) 67.6 | (k) 26.4   | (l) 68.5 | (m) 25.3   | (n) 69.4 | (o) 23.9   | (p) 70.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 52.2   | 58.2  | 58.9  | 56.8  | 54.1  | 51.2  | 47.3  | 46.3  | 47.2  | 48.7  | 50.2  | 52.5  | 55.8  |
|                                                                        | DBStd       | 5.56   | 3.73  | 3.50  | 3.62  | 3.49  | 4.03  | 3.63  | 3.57  | 3.42  | 3.25  | 3.38  | 3.37  | 3.76  |
|                                                                        | HDD50       | 472    | 0     | 0     | 1     | 7     | 33    | 95    | 125   | 99    | 61    | 37    | 12    | 2     |
|                                                                        | HDD65       | 4669   | 214   | 173   | 256   | 328   | 429   | 530   | 580   | 552   | 489   | 458   | 376   | 284   |
|                                                                        | CDD50       | 1284   | 254   | 249   | 211   | 129   | 69    | 15    | 10    | 12    | 22    | 44    | 86    | 183   |
|                                                                        | CDD65       | 5      | 3     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 8      | 6     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 1      | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 6.7    | 6.0   | 5.5   | 5.8   | 6.2   | 6.8   | 7.2   | 7.4   | 6.8   | 7.1   | 7.5   | 7.2   | 6.5   |
| (15) Precipitation                                                     | PrecAvg     | 148.4  | 12.0  | 9.3   | 11.5  | 10.6  | 13.0  | 13.7  | 10.6  | 11.7  | 13.6  | 14.5  | 12.6  | 14.8  |
|                                                                        | PrecMax     | 190.2  | 24.1  | 17.2  | 23.8  | 19.3  | 26.6  | 25.2  | 18.0  | 23.3  | 27.2  | 24.6  | 21.7  | 31.1  |
|                                                                        | PrecMin     | 128.4  | 2.8   | 2.8   | 5.0   | 3.7   | 5.3   | 6.1   | 3.8   | 5.8   | 3.9   | 7.7   | 3.0   | 7.8   |
|                                                                        | PrecStd     | 16.2   | 4.8   | 3.9   | 4.5   | 4.7   | 4.2   | 5.0   | 3.6   | 4.1   | 5.3   | 4.3   | 4.8   | 5.3   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 72.6  | 71.1  | 69.3  | 65.6  | 63.2  | 58.5  | 57.8  | 59.3  | 60.5  | 64.6  | 66.1  | 69.2  |
|                                                                        |             | MCWB   | 63.6  | 63.7  | 61.0  | 57.4  | 55.5  | 52.5  | 48.4  | 48.7  | 50.4  | 51.9  | 58.6  | 63.1  |
|                                                                        | 2%          | DB     | 69.1  | 69.1  | 66.9  | 63.2  | 60.4  | 56.7  | 55.8  | 56.3  | 57.8  | 60.3  | 62.7  | 66.5  |
|                                                                        |             | MCWB   | 63.7  | 63.5  | 61.2  | 57.7  | 55.5  | 53.2  | 49.5  | 49.8  | 51.9  | 53.3  | 57.1  | 61.5  |
|                                                                        | 5%          | DB     | 67.2  | 67.5  | 65.5  | 61.8  | 59.0  | 55.2  | 54.2  | 54.7  | 56.5  | 58.3  | 61.0  | 64.7  |
|                                                                        |             | MCWB   | 62.3  | 62.6  | 60.7  | 57.2  | 55.4  | 51.6  | 49.0  | 50.0  | 51.9  | 53.5  | 56.2  | 60.3  |
|                                                                        | 10%         | DB     | 65.6  | 66.1  | 64.0  | 60.6  | 57.7  | 53.7  | 52.7  | 53.5  | 55.3  | 56.9  | 59.5  | 63.0  |
|                                                                        |             | MCWB   | 61.3  | 61.9  | 59.7  | 56.7  | 54.5  | 49.9  | 48.0  | 49.4  | 51.5  | 52.9  | 55.5  | 59.3  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 67.4  | 66.6  | 65.6  | 61.9  | 59.6  | 56.7  | 54.8  | 54.4  | 55.6  | 57.5  | 61.1  | 64.5  |
|                                                                        |             | MCDB   | 69.9  | 68.5  | 66.7  | 63.4  | 60.6  | 57.3  | 55.6  | 55.3  | 56.6  | 59.4  | 63.8  | 67.4  |
|                                                                        | 2%          | WB     | 64.9  | 65.2  | 63.8  | 60.3  | 58.0  | 54.9  | 53.0  | 52.9  | 54.5  | 56.1  | 59.0  | 62.8  |
|                                                                        |             | MCDB   | 67.4  | 67.3  | 65.3  | 61.4  | 59.1  | 56.0  | 54.3  | 54.3  | 55.9  | 57.7  | 60.8  | 65.2  |
|                                                                        | 5%          | WB     | 63.5  | 64.0  | 62.2  | 59.0  | 56.7  | 53.1  | 51.2  | 51.5  | 53.5  | 55.1  | 57.6  | 61.4  |
|                                                                        |             | MCDB   | 65.9  | 66.4  | 64.1  | 60.5  | 58.1  | 54.4  | 52.8  | 53.5  | 55.4  | 56.7  | 59.7  | 63.7  |
|                                                                        | 10%         | WB     | 62.3  | 62.8  | 60.8  | 57.6  | 55.1  | 51.1  | 49.3  | 50.2  | 52.4  | 53.9  | 56.4  | 60.1  |
|                                                                        |             | MCDB   | 64.7  | 65.4  | 63.2  | 59.7  | 57.1  | 52.7  | 51.3  | 52.7  | 54.6  | 56.1  | 58.6  | 62.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.9  | 12.4  | 12.4  | 12.0  | 10.9  | 11.1  | 11.6  | 11.8  | 12.1  | 12.2  | 12.0  | 11.4  |
|                                                                        |             | MCDBR  | 14.1  | 13.9  | 13.4  | 12.7  | 10.8  | 11.5  | 12.0  | 12.4  | 13.6  | 13.7  | 14.4  | 13.1  |
|                                                                        |             | MCWBR  | 10.0  | 9.9   | 9.6   | 9.3   | 8.2   | 8.9   | 8.6   | 9.0   | 10.0  | 9.7   | 10.6  | 9.5   |
|                                                                        | 5% WB       | MCDBR  | 11.3  | 11.4  | 10.3  | 10.5  | 8.9   | 9.1   | 10.0  | 10.7  | 10.8  | 10.1  | 11.5  | 10.5  |
|                                                                        |             | MCWBR  | 8.8   | 9.0   | 8.7   | 9.2   | 8.1   | 8.6   | 8.8   | 8.9   | 9.1   | 8.7   | 9.2   | 8.3   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.347 | 0.337 | 0.320 | 0.312 | 0.298 | 0.293 | 0.284 | 0.301 | 0.337 | 0.340 | 0.337 | 0.339 |
|                                                                        | taud        |        | 2.535 | 2.582 | 2.631 | 2.633 | 2.643 | 2.601 | 2.627 | 2.610 | 2.489 | 2.493 | 2.526 | 2.558 |
|                                                                        | Ebn at Noon |        | 310   | 304   | 295   | 274   | 256   | 243   | 258   | 274   | 284   | 299   | 311   | 314   |
|                                                                        | Edh at Noon |        | 34    | 30    | 26    | 23    | 18    | 17    | 18    | 22    | 30    | 33    | 34    | 34    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1896  | 1718  | 1273  | 855   | 508   | 411   | 480   | 707   | 1048  | 1424  | 1814  | 1958  |
|                                                                        | RadStd      |        | 142   | 143   | 83    | 81    | 42    | 42    | 43    | 58    | 104   | 119   | 175   | 129   |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A   | N/A     | +0.97  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page